

Joint Confidence Regions for Rankings based on Correlated Estimates

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Background of the Study

- In the problem of estimating ranks of several unknown real-valued parameters $\theta_1, \theta_2, \dots, \theta_K$, it is desired to rank these K parameters from smallest to largest, $\theta_{(1)} < \theta_{(2)} < \dots < \theta_{(K)}$.
- Because rankings based on $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ can change due to sampling variability, statements of uncertainty should accompany released rankings.
- The margin of error gives uncertainty in $\hat{\theta}_k$ for each k separately.
- A direct assessment of the uncertainty in the estimated overall ranking would simultaneously involve all units.

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- Because rankings based on $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ can change due to sampling variability, statements of uncertainty should accompany released rankings.
- The margin of error gives uncertainty in $\hat{\theta}_k$ for each k separately.
- A direct assessment of the uncertainty in the estimated overall ranking would simultaneously involve all units.

- Example 30 day mortality rate in hospitals, mean travel time to work by Klein
- Such tables motivate "implicit" rankings.
- Because rankings based on the observed values of $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ can vary because of sampling variability, widely understood statements of uncertainty should accompany each released ranking.
- While the margin of error gives uncertainty in the estimate $\hat{\theta}_k$ for each unit k separately, A direct assessment of the uncertainty in the estimated overall ranking would jointly involve all units and their relative standing to each other.
- If we repeat the entire process of generating all K intervals many times, at least 95 percent of the time, all K of the resulting intervals will simultaneously contain their respective true parameters $\theta_1, \dots, \theta_K$

Background of the Study

- Let $r_1, r_2, \dots, r_K$ be the true unknown ranks of $\theta_1, \theta_2, \dots, \theta_K$. A mathematical definition of r_k is as follows:

$$r_k = \sum_{j=1}^K I(\theta_j \leq \theta_k) = 1 + \sum_{j:j \neq k} I(\theta_j \leq \theta_k), \quad \text{for } k = 1, 2, \dots, K.$$

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Background of the Study

- the k th rank is the count of parameters, including itself, that are less than or equal to the k th parameter.

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Klein's Approach

Klein et al. (2020) assumes that $\hat{\theta}_k \sim N(\theta_k, \sigma_k^2), k = 1, 2, \dots, K$ where θ_k is unknown but σ_k^2 is known.

Suppose that for each $k \in \{1, 2, \dots, K\}$ there exists values L_k and U_k ST

$$\theta_k \in (L_k, U_k), k = 1, 2, \dots, K, \tag{1}$$

the range for r_k for each $k \in \{1, 2, \dots, K\}$ is as follows:

$$r_k \in \{|\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, |\Lambda_{Lk}| + 3, \dots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1\} \tag{2}$$

where $I_k = \{1, 2, \dots, K\} \setminus \{k\}$

$$\Lambda_{Lk} = \{j \in I_k \mid U_j \leq L_k\}$$

$$\Lambda_{Ok} = \{j \in I_k \mid U_j > L_k \text{ and } U_k > L_j\}$$

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$$\Lambda_{Lk} = \{j \in I_k \mid U_j \leq L_k\}$$

$$\Lambda_{Ok} = \{j \in I_k \mid U_j > L_k \text{ and } U_k > L_j\}$$

Klein's Approach

k	$\hat{\theta}_k$	$\hat{r}_k$	(L_k, U_k)	$ \Lambda_{Lk} $	$ \Lambda_{Ok} $	range
1	12.3	1	(11.0, 14.3)	0	1	1,2
2	17.2	2	(14.1, 21.2)	0	4	1,2,3,4,5
3	19.1	5	(18.2, 20.1)	2	2	3,4,5
4	18.0	3	(17.3, 18.0)	1	1	2,3
5	19.0	4	(18.4, 21.0)	2	2	3,4,5

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- Note that the number of elements in the range given in (2) is $|\Lambda_{Ok}| + 1$.

Suppose that for random quantities L_k and U_k , the intervals satisfy the following probability condition:

$$P \left[\bigcap_{k=1}^K \{ \theta_k \in (L_k, U_k) \} \right] \geq 1 - \alpha; \quad (3)$$

then, by the result of Klein et al. (2020), it also follows that

$$P \left[\bigcap_{k=1}^K \{ r_k \in \{ |\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, \dots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1 \} \} \right] \geq 1 - \alpha. \quad (4)$$

└ Klein's Approach

- We want narrower confidence intervals for $\theta_1, \theta_2, \dots, \theta_K$ since smaller difference between U_k and L_k leads to a smaller $|\Lambda_{Ok}|$.

$$P \left[\bigcap_{k=1}^K \{ \theta_k \in (L_k, U_k) \} \right] \geq 1 - \alpha; \quad (3)$$

$$P \left[\bigcap_{k=1}^K \{ r_k \in \{ |\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, \dots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1 \} \} \right] \geq 1 - \alpha. \quad (4)$$

- Due to the assumption of normality on $\hat{\theta}_k$ as well as the fact that σ_k is assumed known, Klein set the confidence intervals (L_k, U_k) for θ_k to be of the form $\hat{\theta}_k \pm t \times \sigma_k$ for $k \in \{1, 2, \dots, K\}$.
- Bonferroni approach: $t = z_{\alpha/2K}$
- Independence assumption: $t = z_{\gamma/2}$ where $\gamma = 1 - (1 - \alpha)^{1/K}$

└ Klein's Approach

- One may use the Bonferroni approach to choose t . If such an approach is used, the choice of t that would satisfy (3) is $t = z_{\alpha/2K}$.
- Another choice of t is one that exploits the independence assumption on $\hat{\theta}_k$. Such a choice is given by $z_{\gamma/2}$ where $\gamma = 1 - (1 - \alpha)^{1/K}$

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- Problem: Assuming independence when constructing joint confidence regions for estimators that are, in fact, correlated may lead to overly conservative and thus wider intervals, implying greater uncertainty.
- Aim: develop a procedure capable of handling such dependencies while maintaining coverage close to the nominal level and producing relatively narrow joint confidence intervals.

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Joint Confidence Regions for Rankings

└ Motivation

Motivation

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Political setting — David & Legara (2015)

- In weak-party systems, candidates who belong to the same political alliance or ticket commonly co-occur in ballots.

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Joint Confidence Regions for Rankings

└ Motivation

- Candidates with a name-recall advantage, such as media celebrities, incumbents, and members of dynastic families
- In the Philippine setting, candidates running under the same alliance often share campaign machinery and voter bases, which induces correlation in their vote totals across districts.

Objective

This research aims to do the following:

- Develop a procedure to construct joint confidence intervals for the ranks and the ranked parameters when the estimates to be ranked may be correlated.
- Evaluate the performance of the proposed approaches under various parameter settings.
- Compare the results of Algorithm 1 to Klein's joint confidence regions that uses Bonferroni and independence assumption, in terms of coverage and tightness measures.
- Apply the proposed approaches to a real-life example.

This research aims to do the following:

- Develop a procedure to construct joint confidence intervals for the ranks and the ranked parameters when the estimates to be ranked may be correlated.
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- Apply the proposed approaches to a real-life example.

- For these use cases,

Definitions and Assumptions

- Define $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K)'$ and assume that $\hat{\theta} \sim N(\theta, \Sigma)$ where $\theta = (\theta_1, \theta_2, \dots, \theta_K)'$ is unknown and Σ is a known $K \times K$ positive definite matrix.
- Note that in the literature on inferences on the ranks, it is customary to assume that the variances are known.
- We express Σ , where R is the correlation matrix.

$$\Sigma = \Delta^{1/2} R \Delta^{1/2}; \quad \Delta = \text{diag} \left\{ \sigma_1^2, \sigma_2^2, \dots, \sigma_K^2 \right\}.$$

Joint Confidence Regions for Rankings

Definitions and Assumptions

- The estimators may be correlated, so the covariance matrix is not necessarily diagonal. We write the covariance matrix in terms of variances and a correlation matrix.
- The diagonal elements of Σ , which are $\sigma_k^2 = V(\hat{\theta}_k)$ for $k = 1, 2, \dots, K$, are treated as known quantities in practice.

- Define $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K)'$ and assume that $\hat{\theta} \sim N(\theta, \Sigma)$ where $\theta = (\theta_1, \theta_2, \dots, \theta_K)'$ is unknown and Σ is a known $K \times K$ positive definite matrix.
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└ Procedure

- 1 Derive simultaneous confidence intervals for $\theta_1, \theta_2, \dots, \theta_K$ of the form

$$\mathfrak{R}_1 = [\hat{\theta}_1 \pm t \times \sigma_1] \times [\hat{\theta}_2 \pm t \times \sigma_2] \times \dots \times [\hat{\theta}_K \pm t \times \sigma_K]. \quad (5)$$

- 2 Use the result of Klein et al. (2020) to get the lower and upper bounds on the ranks $r_k, k = 1, 2, \dots, K$.

- bullet 2: Once the confidence intervals in have been obtained, we can then use the result of Klein to get the lower and upper bounds on the ranks $r_k, k = 1, 2, \dots, K$. That is, we also get a joint confidence region for $r_1, r_2, \dots, r_K$.

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- 2 Use the result of Klein et al. (2020) to get the lower and upper bounds on the ranks $r_k, k = 1, 2, \dots, K$.

Proposed methodology to compute the joint confidence region for the unordered parameters: Algorithm 1

Let the data be represented by $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K)'$ and suppose that Σ is known

1: **for** $b = 1, 2, \dots, B$ **do**

2: Generate $\hat{\theta}_b^* \sim N_K(\hat{\theta}, \Sigma)$ and write $\hat{\theta}_b^* = (\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*)'$

3: Compute

$$t_b^* = \max_{1 \leq k \leq K} \left| \frac{\hat{\theta}_{bk}^* - \hat{\theta}_k}{\sigma_k} \right|$$

4: **end for**

5: Compute the $(1 - \alpha)$ -sample quantile of $t_1^*, t_2^*, \dots, t_B^*$, call this $\hat{t}$.

6: The joint confidence region for $\theta = (\theta_1, \theta_2, \dots, \theta_K)'$ is given by

$$\mathfrak{R}_1 = [\hat{\theta}_1 \pm \hat{t} \times \sigma_1] \times [\hat{\theta}_2 \pm \hat{t} \times \sigma_2] \times \dots \times [\hat{\theta}_K \pm \hat{t} \times \sigma_K].$$

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Proposed methodology to compute the joint confidence region for the unordered parameters:

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Remarks about bootstrap (Hesterberg, 2021):

- We plugin the estimator of an unknown parameter to the normal distribution
- Bootstrap is implemented by Monte Carlo sampling.
- We estimate the distribution of t directly from the data, contrary to using the z-table
- Bootstrap distribution is centered at the observed statistic, not the parameter. That is, the mean of the bootstrap estimates is not μ but $\bar{x}$. As a result, we do not use the mean of the bootstrap statistic as a replacement for the original estimator (e.g., bootstrap cannot improve upon $\bar{x}$) as an estimator for μ .
- Bootstrap is for estimating only the SE (it is only use to estimate parameter for some, e.g. quantiles)

Proposed methodology to compute a joint confidence region for the ordered parameters

- 1: **for** $b = 1, 2, \dots, B$ **do**
- 2: Generate $\hat{\boldsymbol{\theta}}_b^* = (\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*)' \sim N_K(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma})$ and let $\hat{\theta}_{b(1)}^*, \hat{\theta}_{b(2)}^*, \dots, \hat{\theta}_{b(K)}^*$ be the corresponding ordered values
- 3: Compute $\hat{\sigma}_{b(k)}^*$ using:
 - asymptotic variance definition (Algorithm 2)
 - second-level bootstrap (Algorithm 3)
- 4: Compute

$$t_b^* = \max_{1 \leq k \leq K} \left| \frac{\hat{\theta}_{b(k)}^* - \hat{\theta}_{(k)}}{\hat{\sigma}_{b(k)}^*} \right|$$

- 5: **end for**
- 6: Compute the $(1 - \alpha)$ -sample quantile of $t_1^*, \dots, t_B^*$, call this $\hat{t}$.
- 7: The joint confidence region of $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$ is given by
$$\mathfrak{R}_2 = \left[\hat{\theta}_{(1)} \pm \hat{t} \times \hat{\sigma}_{(1)} \right] \times \left[\hat{\theta}_{(2)} \pm \hat{t} \times \hat{\sigma}_{(2)} \right] \times \dots \times \left[\hat{\theta}_{(K)} \pm \hat{t} \times \hat{\sigma}_{(K)} \right]$$

Joint Confidence Regions for Rankings

└ Proposed methodology to compute a joint confidence region for the ordered parameters

- We can say we are 95 percent confident that the smallest parameter up to the largest parameter falls between the calculated bounds.

Proposed methodology to compute a joint confidence region for the ordered parameters

```
1: for  $b = 1, 2, \dots, B$  do
2:   Generate  $\hat{\boldsymbol{\theta}}_b^* = (\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*)' \sim N_K(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma})$  and let  $\hat{\theta}_{b(1)}^*, \hat{\theta}_{b(2)}^*, \dots, \hat{\theta}_{b(K)}^*$  be the corresponding ordered values
3:   Compute  $\hat{\sigma}_{b(k)}^*$  using:
     • asymptotic variance definition (Algorithm 2)
     • second-level bootstrap (Algorithm 3)
4:   Compute  $t_b^* = \max_{1 \leq k \leq K} \left| \frac{\hat{\theta}_{b(k)}^* - \hat{\theta}_{(k)}}{\hat{\sigma}_{b(k)}^*} \right|$ 
5: end for
6: Compute the  $(1 - \alpha)$ -sample quantile of  $t_1^*, \dots, t_B^*$ , call this  $\hat{t}$ .
7: The joint confidence region of  $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$  is given by
    $\mathfrak{R}_2 = \left[ \hat{\theta}_{(1)} \pm \hat{t} \times \hat{\sigma}_{(1)} \right] \times \left[ \hat{\theta}_{(2)} \pm \hat{t} \times \hat{\sigma}_{(2)} \right] \times \dots \times \left[ \hat{\theta}_{(K)} \pm \hat{t} \times \hat{\sigma}_{(K)} \right]$ 
```

Algorithm 2: Asymptotic Definition of Variance

- This uses results from Chen (1976) and Dudewicz (1972) to obtain an expression of the asymptotic variance of $\hat{\theta}_{(k)}$

$$\hat{\sigma}_{b(k)}^* = \sqrt{\left[\text{kth ordered value among } \left\{ \hat{\theta}_{b1}^{*2} + \sigma_1^2, \dots, \hat{\theta}_{bK}^{*2} + \sigma_K^2 \right\} \right] - \hat{\theta}_{(k)}^{*2}}$$

- In $\mathfrak{R}_2$, $\hat{\sigma}_{(k)} =$

$$\sqrt{\text{kth ordered value among } \left\{ \hat{\theta}_1^2 + \sigma_1^2, \hat{\theta}_2^2 + \sigma_2^2, \dots, \hat{\theta}_K^2 + \sigma_K^2 \right\} - \hat{\theta}_{(k)}^2}$$

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Algorithm 3: Variance from Second-Level Bootstrap

• Second-level bootstrap algorithm

- 1: **for** $c = 1, 2, \dots, C$ **do**
- 2: Generate $\hat{\theta}_{bc}^{**} = (\hat{\theta}_{bc1}^{**}, \hat{\theta}_{bc2}^{**}, \dots, \hat{\theta}_{bcK}^{**}) \sim N_K(\hat{\theta}_b^*, \Sigma)$ and let $\hat{\theta}_{bc(1)}^{**}, \hat{\theta}_{bc(2)}^{**}, \dots, \hat{\theta}_{bc(K)}^{**}$ be the corresponding ordered values of $\hat{\theta}_{bc1}^{**}, \hat{\theta}_{bc2}^{**}, \dots, \hat{\theta}_{bcK}^{**}$
- 3: Compute $\hat{\sigma}_{b(k)}^* = \sqrt{\frac{\sum_{c=1}^C (\hat{\theta}_{bc(k)}^{**} - \bar{\theta}_{b \cdot (k)}^{**})^2}{C - 1}}$, $\bar{\theta}_{b \cdot (k)}^{**} = \frac{1}{C} \sum_{c=1}^C \hat{\theta}_{bc(k)}^{**}$
- 4: **end for**

$$\text{In } \mathfrak{R}_2, \hat{\sigma}_{(k)} = \sqrt{\frac{\sum_{b=1}^B (\hat{\theta}_{b(k)}^* - \bar{\theta}_{\cdot(k)}^*)^2}{B - 1}}, \quad \bar{\theta}_{\cdot(k)}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_{b(k)}^*$$

Joint Confidence Regions for Rankings

Algorithm 3: Variance from Second-Level Bootstrap

- unavailable SE formula - we would need to compute a bootstrap estimate for the standard error for each bootstrap sample.
- involves taking second level, nested bs samples that will be used to estimate se.
-

Algorithm 3: Variance from Second-Level Bootstrap

```

1: Second-level bootstrap algorithm
2: for c = 1, 2, ..., C do
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4:   Compute  $\hat{\sigma}_{b(k)}^* = \sqrt{\frac{\sum_{c=1}^C (\hat{\theta}_{bc(k)}^{**} - \bar{\theta}_{b \cdot (k)}^{**})^2}{C - 1}}$ ,  $\bar{\theta}_{b \cdot (k)}^{**} = \frac{1}{C} \sum_{c=1}^C \hat{\theta}_{bc(k)}^{**}$ 
5: end for
6: In  $\mathfrak{R}_2$ ,  $\hat{\sigma}_{(k)} = \sqrt{\frac{\sum_{b=1}^B (\hat{\theta}_{b(k)}^* - \bar{\theta}_{\cdot(k)}^*)^2}{B - 1}}$ ,  $\bar{\theta}_{\cdot(k)}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_{b(k)}^*$ 

```

Evaluation Algorithm

For given values of Σ and $\theta_1, \theta_2, \dots, \theta_K$ (with corresponding $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$ for ordered parameters)

- 1: **for** replications = 1, 2, ..., 5000 **do**
- 2: Generate $\hat{\theta} \sim N_K(\theta, \Sigma)$
- 3: Compute the confidence region $\mathfrak{R}_1$ for the unordered parameters using Algorithm 1 and the confidence region for the ordered parameters $\mathfrak{R}_2$ using Algorithms 2 and 3.
- 4: For the unordered parameters, check if $(\theta_1, \theta_2, \dots, \theta_K) \in \mathfrak{R}_1$ and compute T_1, T_2 , and T_3 . For the ordered parameters, check if $(\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}) \in \mathfrak{R}_2$
- 5: **end for**
- 6: Compute the proportion of times that the condition in line 4 is satisfied and the average of T_1, T_2 , and T_3 .

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$$T_1 = \frac{1}{K} \sum_{k=1}^K |\Lambda_{Ok}|$$

$$T_2 = \prod_{k=1}^K |\Lambda_{Ok}|$$

$$T_3 = 1 - \frac{OP}{K^2}; OP = K + \sum_{k=1}^K |\Lambda_{Ok}|$$

Measures of tightness

- Moreover, the tightness of the joint confidence region that results from Algorithm 1 is assessed using three summary measures: the arithmetic mean (T_1), geometric mean (T_2), and the metric T_3 introduced by Wright (2025)
- OP denotes the total number of occupied positions in a joint confidence region out of the total number of positions K^2 ; or the sum of the differences between the upper and lower bound of the simultaneous rank intervals added by 1, for each population k .
- Higher values of T_1 and T_2 indicate wider confidence intervals and are therefore less desirable, whereas higher values of T_3 are preferable. T_3 can range from 0, indicating no tightness, to $\frac{K-1}{K}$, implying the confidence region only contains the estimated ranking which is likely the true ranking.

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$$T_2 = \prod_{k=1}^K |\Lambda_{Ok}|$$

$$T_3 = 1 - \frac{OP}{K^2}, OP = K + \sum_{k=1}^K |\Lambda_{Ok}|$$

Simulation settings

The following settings are considered for $K = \{10, 20, 30, 40, 50\}$. Each K has a set of corresponding variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_K^2$. A nominal level of $1 - \alpha = 0.95$ will be used.

θ	Correlation matrix
low variability	$\mathbf{R}_{eq}, \mathbf{R}_{bl,2}, \mathbf{R}_{bl,3}$
medium variability	$\mathbf{R}_{eq}, \mathbf{R}_{bl,2}, \mathbf{R}_{bl,3}$
high variability	$\mathbf{R}_{eq}, \mathbf{R}_{bl,2}, \mathbf{R}_{bl,3}$

- We assume certain correlation structures among the $\hat{\theta}s$.
- Equicorrelation is considered for simplicity.

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Correlation Structures

- This assumes that the k variables are equally correlated, i.e., that $\rho_{jk} = \rho$ where $\rho \in [-1, 1]$ for $j \neq k \in \{1, \dots, K\}$. For the simulation we let $\rho \in \{0.1, 0.5, 0.9\}$.

$$\mathbf{R}_{eq} = (1 - \rho) \mathbf{I}_K + \rho \mathbf{1}_K \mathbf{1}_K' = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}_{K \times K}$$

- The block diagonal correlation matrix can be expressed as

$$\mathbf{R}_{bl,G} = \begin{bmatrix} \mathbf{R}_{eq,1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_{eq,2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{R}_{eq,G} \end{bmatrix}_{K \times K}$$

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Table 1: Configuration for $\mathbf{R}_{bl,G}$

G = 2				G = 3			
R	g	Size	ρ_g	R	g	Size	ρ_g
$\mathbf{R}_{bl,2B}$	1	0.5K x 0.5K	0.9	$\mathbf{R}_{bl,3L}$	1	0.2K x 0.2K	0.9
	2	0.5K x 0.5K	0.1		2	0.3K x 0.3K	0.5
$\mathbf{R}_{bl,2U}$	1	0.3K x 0.3K	0.9		3	0.5K x 0.5K	0.1
	2	0.7K x 0.7K	0.1	$\mathbf{R}_{bl,3H}$	1	0.2K x 0.2K	0.1
					2	0.3K x 0.3K	0.5
					3	0.5K x 0.5K	0.9

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Correlation Structures

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- Useful in the context of pre-election surveys
- In a block correlation matrix $\mathbf{R}_{block}$ with G blocks, each diagonal block represents an equicorrelation structure within group g , denoted by

$$\mathbf{R}_{eq,g} = (1 - \rho_g) \mathbf{I}_{n_g} + \rho_g \mathbf{1}_{n_g} \mathbf{1}_{n_g}'$$

where ρ_g is the within-block correlation and n_g is the number of variables in block g such that $\sum_{g=1}^G n_g = K$.

- The off-diagonal blocks capture between-block correlations, represented by $\mathbf{C}_{g'g} = \mathbf{C}_{gg'} = \rho_{gg'} \mathbf{1}_{n_g} \mathbf{1}_{n_{g'}}'$ where $g \neq g' \in \{1, \dots, G\}$

Table 2: Simulation Results for Coverage of Unordered Parameters

Equicorrelated				Block Diagonal			
	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$		$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$
K = 10							
$\mathbf{R}_{eq,0.1}$	0.947	0.948	0.946	$\mathbf{R}_{bl,3L}$	0.955	0.957	0.950
$\mathbf{R}_{eq,0.5}$	0.961	0.962	0.950	$\mathbf{R}_{bl,2U}$	0.957	0.958	0.951
$\mathbf{R}_{eq,0.9}$	0.984	0.984	0.948	$\mathbf{R}_{bl,2B}$	0.962	0.963	0.949
				$\mathbf{R}_{bl,3H}$	0.963	0.964	0.949
K = 20							
$\mathbf{R}_{eq,0.1}$	0.945	0.947	0.944	$\mathbf{R}_{bl,3L}$	0.954	0.955	0.948
$\mathbf{R}_{eq,0.5}$	0.961	0.961	0.946	$\mathbf{R}_{bl,2U}$	0.955	0.957	0.944
$\mathbf{R}_{eq,0.9}$	0.986	0.986	0.952	$\mathbf{R}_{bl,2B}$	0.966	0.967	0.946
				$\mathbf{R}_{bl,3H}$	0.966	0.967	0.949

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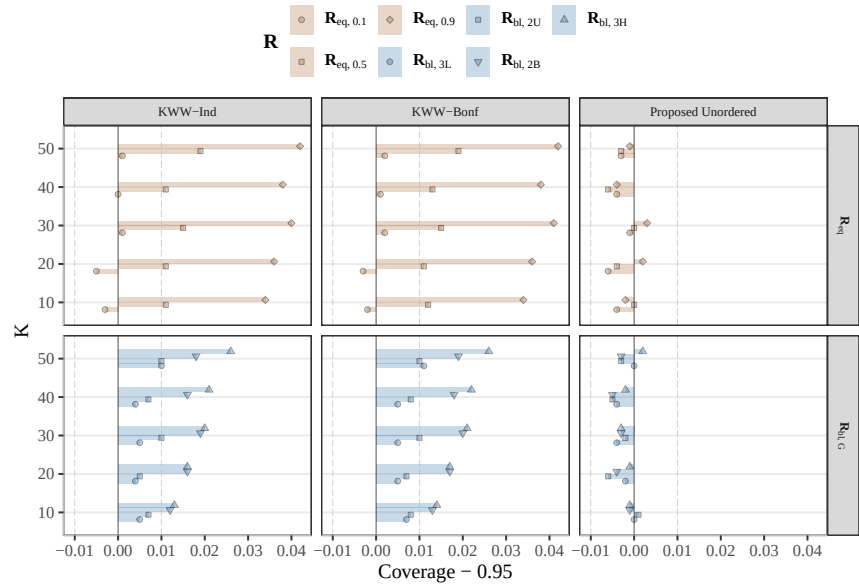
Coverage of Unordered Parameters

Coverage of Unordered Parameters

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	Equicorrelated			Block Diagonal			
	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	
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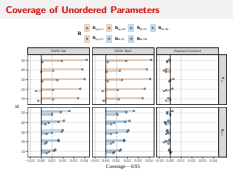
Coverage of Unordered Parameters



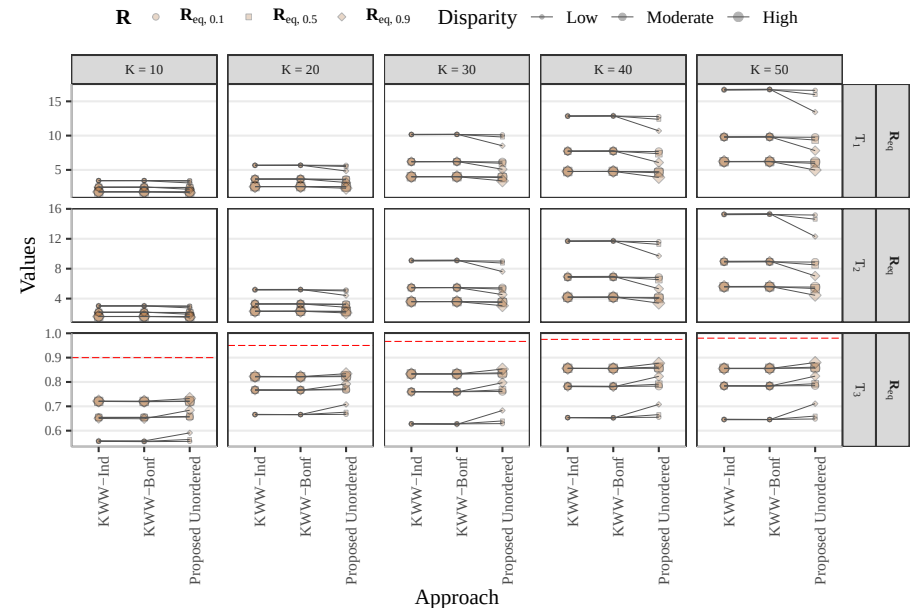
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Joint Confidence Regions for Rankings

└ Coverage of Unordered Parameters



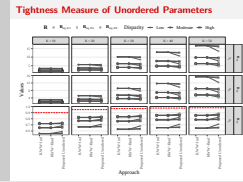
Tightness Measure of Unordered Parameters



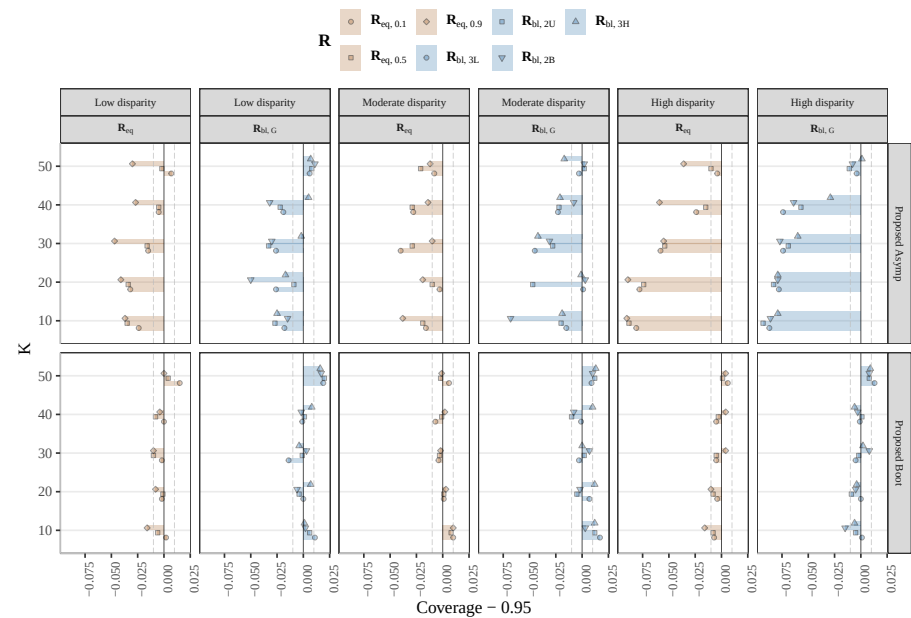
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Joint Confidence Regions for Rankings

Tightness Measure of Unordered Parameters



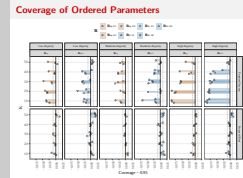
Coverage of Ordered Parameters



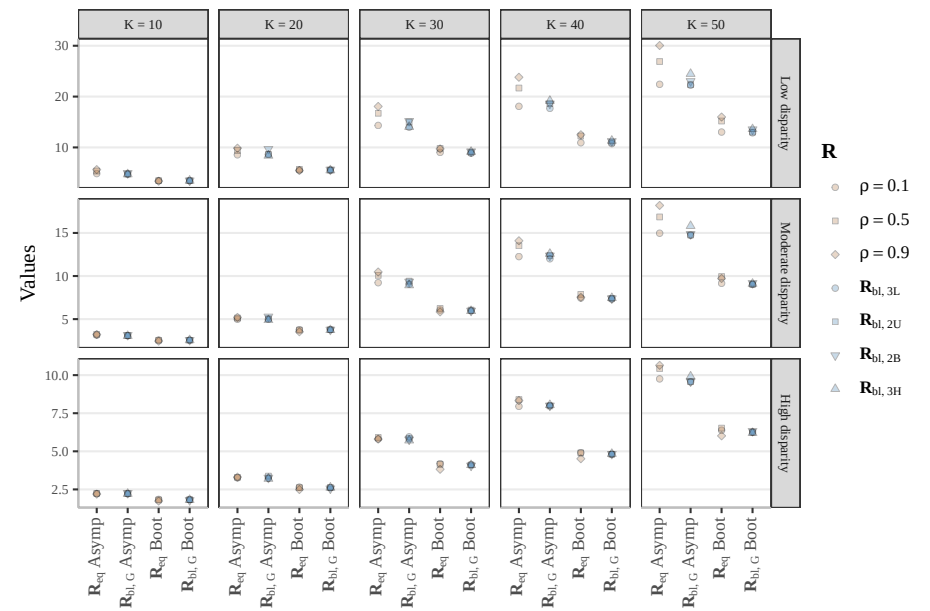
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Joint Confidence Regions for Rankings

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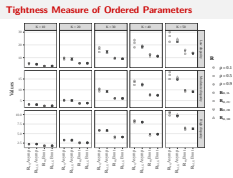
Tightness Measure of Ordered Parameters



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Joint Confidence Regions for Rankings

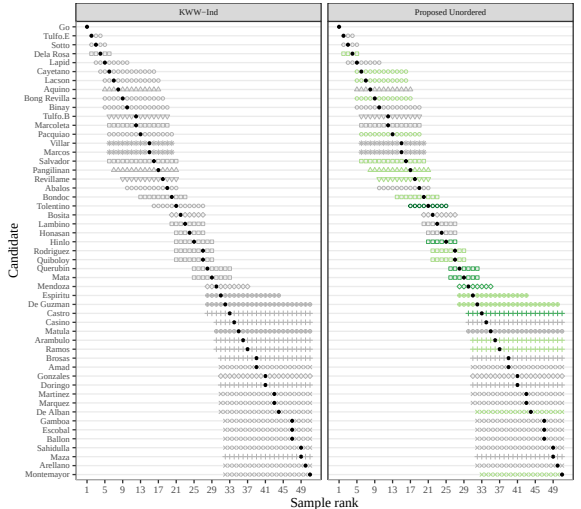
Tightness Measure of Ordered Parameters



Application

o Alyansa * DuterTen-Alyansa • Labor ▼ Media ◇ Traditional
□ DuterTen △ KiBam + Makabayan × Sectoral

Decrease in Occupied Rank Position 0 1 2 3

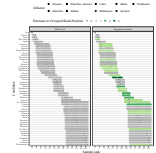


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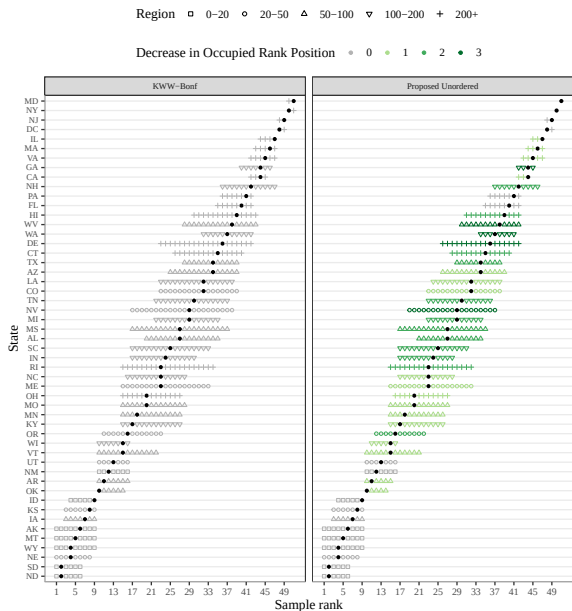
Joint Confidence Regions for Rankings

└ Application

Application



Application



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Application

